

Packet Tracer - Observe Traffic Flow in a Routed Network

Objectives

Part 1: Observe Traffic Flow in an Unrouted LAN

Part 2: Reconfigure the Network to Route Between LANs

Part 3: Observe Traffic Flow in the Routed Network

Background / Scenario

The company that you work for has been asked to propose a new network design for XYZ LLC. XYZ is a startup company that has recently experienced success with their product offerings. They will be expanding, and their network will need to grow with them. Currently, the network is configured with a single IP network for hosts in all departments. This design has become inefficient and network delays are becoming increasingly noticeable. You have been asked to help prepare the proposal with the sales team. The sales team will propose a solution in which network efficiency is enhanced by implementing routing between separate department networks. You are working on a demonstration of how having multiple routed networks in a business can improve network efficiency. Follow the instructions to go through the demonstration to help propose a new network to XYZ LLC.

Instructions

Part 1: Observe Traffic Flow in an Unrouted LAN

The XYZ network consists of about 150 devices that are connected to a LAN. The LAN is configured on a single IPv4 network. Hosts in different departments are connect to switches which are then connected to the **Edge** router. The router only routes traffic between the LAN and the internet, represented by the **ISP** cloud. Because only one IP network is used in the LAN, all departments are on the same network.

The Packet Tracer topology is simplified. It only shows some of the departments and hosts. Assume that the behavior that you will demonstrate is happening at far greater scale than what is shown in the PT network.

In this part, you will use Packet Tracer simulation mode to observe how traffic flows through an unrouted LANs.

Step 1: Clear the ARP cache on host Sales 1.

Hover your mouse over the **Sales 1** host to see its IP address. Make note of it.

- Click **Sales 1 > Desktop** tab > **Command Prompt**, and then enter the **arp -a** command. There should be no MAC addresses in the ARP cache. If there are entries in the ARP cache, use the **arp -d** command to delete them.

Step 2: Observe traffic flow in the network.

- Click the **Simulation** mode button in the lower right-hand corner of the PT window to switch from **Realtime** to **Simulation** mode.
- Open the **Command Prompt** for **Sales 2**, and then enter the **ping** command followed by the IP address of **Sales 1**.
- Use the **Capture then Forward** button (the triangle pointing to the right with a vertical bar attached) in the **Play Controls** of the **Simulation Panel** to begin to execute the **ping** command. You will see a colored envelope icon appear next to Sales 2. This represents a PDU. Click the **Capture then Forward** button to

move the PDU to the first device on its path to the destination device. Click the PDU envelope to inspect the contents.

What are the source and destination MAC and IP addresses for the frame and packet? **Answer: The frame's src MAC address is sales 1' mac address and the destination MAC address is the broadcast address FFFF.FFFF.FFFF. The packet's source IP is Sale 1's IP address, and the destination IP is Sales 2's.**

Why is the destination MAC address the broadcast address? **Answer: Since the host's ARP cache is empty, it has to send a ARP request first to get the destination MAC address so the frame can be sent to Sales 1.**

- d. Advance the PDUs through the network until a new PDU (different color) is created at **Sales 2**.

Which hosts and other types of devices needed to process the ARP request packets? **Answer: The hosts and router interface.**

What is the impact of this on efficient operation of the network as it is currently configured? **Answer: If the destination is close, an empty ARP cache triggers a broadcast ARP request to all hosts. With many devices, these frequent broadcasts and reduce space for normal traffic.**

- e. A new PDU with a different color has appeared at Sales 2. Click the new PDU and inspect its contents. Look at the outbound PDU details.

What type of PDU is this? **Answer: It's the first ICMP echo- request packet that is issued by a ping from host Sales 2.**

- f. Return to **Realtime** mode.

Part 2: Reconfigure the Network to Route Between LANs.

In this part, you will demonstrate the benefits of routing between department networks. First, you will cable each network switch to connect directly to a router interface. Then, you will reconfigure the hosts to receive addresses on two new IPv4 networks that are created by the router.

Step 1: Change device connections.

The three switches are connected to each other with copper straight through cables.

- For the cable that connects the **Accounting** switch with **Finance** switch, click the green triangle on the **Accounting** switch side of the link.
- Drag that end of the cable to the **Edge** router and connect the cable to port **GigabitEthernet 1/0**.
- Repeat this step for the link between **Finance** and **Sales**. Connect to the available GigabitEthernet port.

Step 2: Configure the hosts with addresses on the new LANs.

Each interface of the **Edge** router was previously configured to put each department on its own IPv4 network. The hosts will receive their new IP addresses from the router. However, it will take time for the hosts on the **Finance** and **Sales** networks to receive their new IP addresses. (The hosts on the Accounting network will remain on 192.168.1.0/24.)

- a. To speed up the process of getting new IP addresses, open a **Command Prompt** on each of the four devices in the **Finance** and **Sales** networks.
- b. Enter the **ipconfig /renew** command. This will force the host to request a new IP address from the DHCP server that is running on the **Edge** router. You should see confirmation of new IP addressing.

What IPv4 network is assigned to the **Finance** network? **Answer: 192.168.2.0/24**

What IPv4 network is assigned to the **Sales** network? **Answer: 192.168.3.0/24**

Part 3: Observe Traffic Flow in the Routed Network.

In this part, you will observe how traffic now flows through a routed network.

Step 1: Ping Sales 1 from Sales 2.

- a. Return to the **Command Prompt** for **Sales 2** and verify that its ARP cache is empty. If it is not, delete any entries.
- b. Switch to **Simulation** mode.
- c. Ping **Sales 1** from **Sales 2**.
- d. Use the **Capture then Forward** button to step the PDUs through the network. Observe how the ARP request message flows through the network this time.

Which devices receive the ARP broadcasts this time? **Answer: Sales 1 and the router interface that is connected to the sales department network process the PDU.**

Step 2: Ping other hosts.

Repeat this demo by pinging other hosts and the internet server. Observe the flow of the ARP request PDUs.

What is the benefit of using multiple IPv4 networks, or subnetworks, within an enterprise? **Answer: A big benefit of using multiple IP networks is that the traffic stays in the areas where it's needed, without slowing down other parts of the network.**

Note: The network topology that is used in the activity is for demonstration purposes only. While it is possible that a real enterprise network could use a router in this way, there are more optimal topologies that achieve these results. You will learn about other design approaches in later networking courses.

Packet Tracer - Observe Traffic Flow in a Routed Network

Cisco Packet Tracer - C:/Users/BrandonJohnson/Downloads/14_3_3_Packet_Tracer_Observe_Traffic_Flow_in_a_Routed_Network (1).pka - Guest - 2026-04-16 23:55:06

File Edit Options View Tools Extensions Window Help

Logical Physical x:517, y:5

XYZ LLC

Accounting
Accounting 1
Accounting 2
Accounting Department

Finance
Finance 1
Finance 2
Finance Department

Sales
Sales 1
Sales 2
Sales Department

PT Activity: 00:26:20

mode.

b. Open the Command Prompt for Sales 2, and then enter the ping command followed by the IP address of Sales 1.

c. Use the Capture then Forward button (the triangle pointing to the right with a vertical bar attached) in the Play Controls of the Simulation Panel to begin to execute the ping command. You will see a colored envelope icon appear next to Sales 2. This represents a PDU. Click the Capture then Forward button to move the PDU to the first device on its path to the destination device. Click the PDU envelope to inspect the contents.

What are the source and destination MAC and IP addresses for the frame and packet?

Why is the destination MAC address the broadcast address?

d. Advance the PDUs through the network until a new PDU (different color) is created at Sales 2.

Which hosts and other types of devices needed to process the ARP request packets?

What is the impact of this on efficient operation of the network as it is currently configured?

e. A new PDU with a different color has appeared at Sales 2. Click the new PDU and inspect its contents. Look at the outbound PDU details.

What type of PDU is this?

f. Return to Realtime mode.

Part 2: Reconfigure the Network to Route Between LANs.

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Step 1: Change device connections.

The three switches are connected to each other with copper straight through cables.

a. For the cable that connects the Accounting switch with Finance switch, click the green triangle on the Accounting switch side of the link.

b. Drag that end of the cable to the Edge router and connect the cable to port GigabitEthernet 1/0.

c. Repeat this step for the link between Finance and Sales. Connect to the available GigabitEthernet port.

Time Elapsed: 00:26:20 Completion: 83%

☐ Top ☐ Dock 1/1

Simulation Panel

Event List

Vis	Time(sec)	Last Device
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Reset Simulation ☒ Constant Delay Captured to (no captures)

Play Controls

Event List Filters - Visible Events

ARP, ICMP